

Lab 8

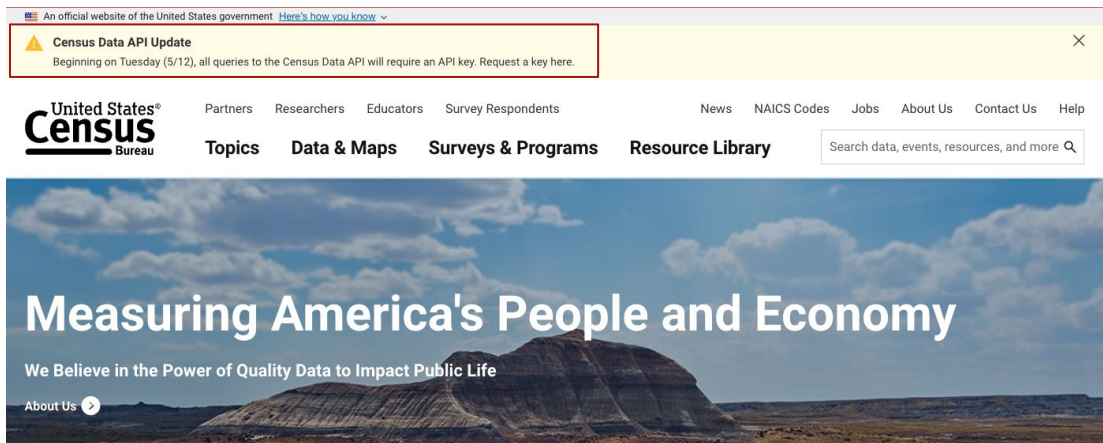
Evictions and Redlining

with data from the Eviction Lab, ACS, NYC Open Data,
& the Mapping Inequality project

DATA 20195 Spring 2026

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Census API Updates



General recommendations moving forward:

- Make sure you have an active Census API key (https://api.census.gov/data/key_signup.html)
- Make any Census request once and then immediately save the data somewhere
- Get any and all Census data needed for your project as soon as possible

Patch for issue with census package (put below code block where you have 'from census import Census'):

```
def _patched_field_type(self, field, year):
    url = self.definition_url % (year, self.dataset, field)
    resp = self.session.get(url, params={"key": self._key})

    type_map = {
        "fips-for": str,
        "fips-in": str,
        "int": float,
        "long": float,
        "float": float,
        "string": str,
    }
    if resp.status_code == 200:
        try:
            predicate_type = resp.json().get("predicateType", "string")
            return type_map.get(predicate_type, str)
        except Exception:
            return str
    return str

# Apply lru_cache and patch onto the base class
Client._field_type = lru_cache(maxsize=1024)(_patched_field_type)
```

Eviction Lab Dataset

Data Curation

- Census (demographic)
- **Eviction records:** collected via court records.
 - **Official court record:** Directly published or requested from civil courts
 - **Proprietary sources:** Electronic and in-person collection of court records
- Despite effort, records are still missing in ~26000 county-years
 - Built model to estimate missing values using primarily official records
 - Proprietary data are used as a noisy measure in the model.

Eviction Lab Dataset

Observed + estimated dataset

	state	county	FIPS_state	FIPS_county	year	renting_hh	filings_estimate	filings_ci_95_lower	filings_ci_95_upper	ind_filings_court_issued	ind_filings_court_issued_LT	hh_threat_estimate	hh_threat_95_lower	hh_threat_95_upper	ind_hht_observed
0	Alabama	Autauga County	1	1001	2000	3074	109	109	109	1	0.0	106	106	106	1
1	Alabama	Autauga County	1	1001	2001	3264	75	75	75	1	0.0	74	71	75	0
2	Alabama	Autauga County	1	1001	2002	3454	94	94	94	1	0.0	92	88	94	0
3	Alabama	Autauga County	1	1001	2003	3644	100	100	100	1	0.0	98	94	100	0
4	Alabama	Autauga County	1	1001	2004	3834	112	112	112	1	0.0	110	106	112	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
195	Alabama	Chilton County	1	1021	2005	3296	67	67	67	1	0.0	66	64	67	0
196	Alabama	Chilton County	1	1021	2006	3412	83	83	83	1	0.0	82	79	83	0
197	Alabama	Chilton County	1	1021	2007	3529	71	71	71	1	0.0	70	68	71	0
198	Alabama	Chilton County	1	1021	2008	3646	39	39	39	1	0.0	39	37	39	0
199	Alabama	Chilton County	1	1021	2009	3763	78	78	78	1	0.0	77	75	78	0

Demographics: county FIPS, year, number of renter-occupied housing units

Evictions Filings: number of evictions filed (a household can have >1 filed eviction)

Count comes directly from court-issued data (1 – yes, 0 – no)

Court-issued data are landlord-tenant cases (1 – yes, 0 – no)

Evictions Filed-on: number of households who received an eviction filing

Count is observed rather than estimated (1 – observed, 0 – estimated)

Eviction Lab Dataset

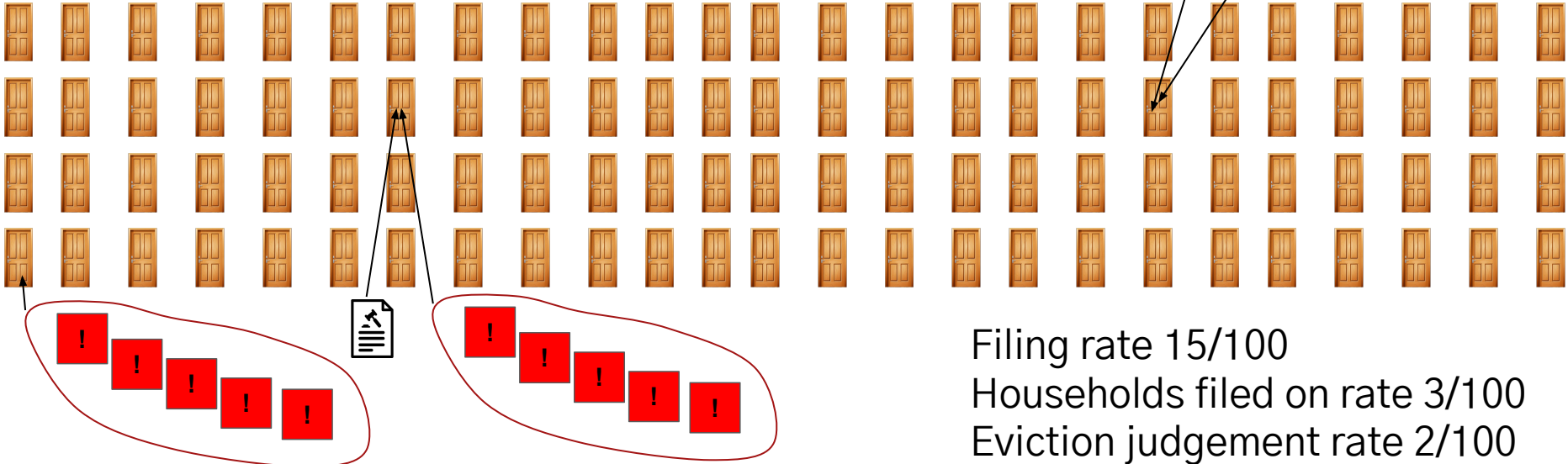
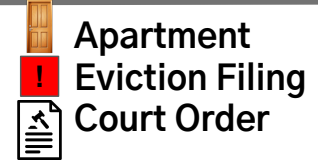
Proprietary dataset

	id	name	parent_location	year	type	filings	filing_rate	hh_filed_on	hh_filed_on_rate	judgements	judgement_rate
0	1001	Autauga County	Alabama	2000	observed	57.0	1.85	54.0	1.76	35.0	1.14
1	1001	Autauga County	Alabama	2001	observed	84.0	2.73	83.0	2.70	35.0	1.14
2	1001	Autauga County	Alabama	2002	observed	100.0	3.25	97.0	3.16	20.0	0.65
3	1001	Autauga County	Alabama	2003	observed	102.0	3.32	90.0	2.93	12.0	0.39
4	1001	Autauga County	Alabama	2004	observed	19.0	0.62	19.0	0.62	12.0	0.39

Eviction Judgement: number of
renters that receive
court-ordered evictions

Building off the provided data

- **Rate of filings:** the total number of evictions *filed* per 100 renter-occupied homes in that area for a given year. Note that this can reflect many filings on one apartment.
- **Households filed on:** the total number of renter-occupied households who ever received an eviction filing in a given year, per 100.
- **Eviction judgment rate:** the number of renters who received an eviction judgment/court order requiring them to leave over all renter-occupied homes in a given year.

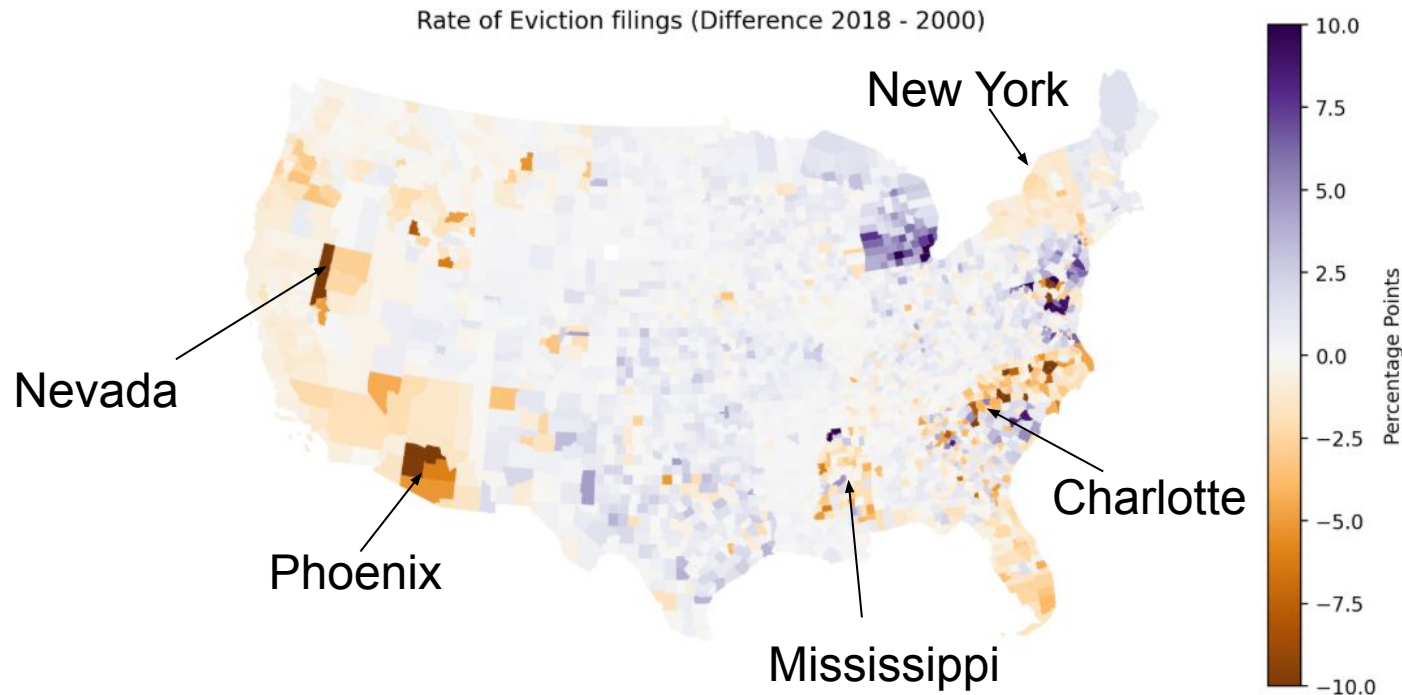


Filing rate 15/100
Households filed on rate 3/100
Eviction judgement rate 2/100

Discussion: Policy traces on the map

Some of the patterns hold for whole states and some show variation below the state level.

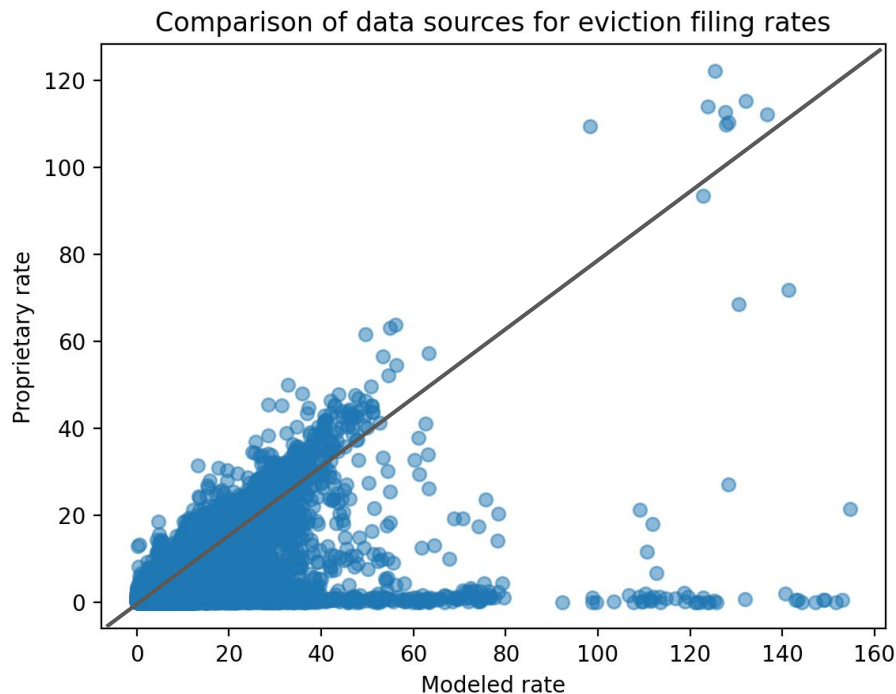
What might be some policy correlates of the patterns?



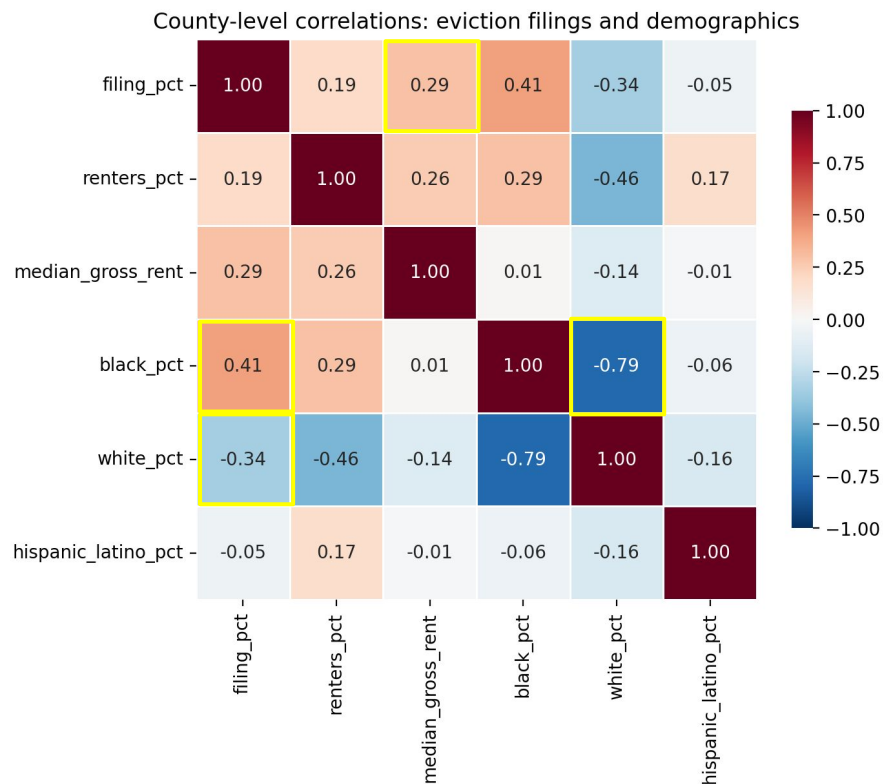
Discussion: Utility of Smoothing

It seems pretty clear that the “modeled” data changes the picture substantially.

Discuss the **advantages of smoothing over anomalies in the data**, and talk about how to do so **responsibly**.



Discussion: Eviction Correlations



A correlation tells you **how reliably two variables move together**, on a scale from -1 to +1.

- **+1** means they move together perfectly
- **-1** means one always goes up when the other goes down
- **0** means knowing one tells you nothing about the other

Correlation does **not** tell you anything about the actual units or magnitude of change.

Surprises?

NYC Eviction Dataset – *executed* evictions

Individual (household/apartment) level eviction records in NYC.

docket_number	eviction_address	eviction_aprt_num	executed_date	marshal_first_name	marshal_last_name	residential_commercial_ind	borough	eviction_zip	ejectment	eviction_possession	latitude	longitude
106626	1347 OCEAN AVENUE	3H	2022-07-28T00:00:00.000	Henry	Daley	Residential	BROOKLYN	11230	Not an Ejectment	Possession	40.629253	-73.957049
110322	134-27 166TH PLACE	12C	2024-02-27T00:00:00.000	Ileana	Rivera	Residential	QUEENS	11434	Not an Ejectment	Possession	40.672889	-73.773607
102136	76 NORTH 4 STREET	422	2020-02-24T00:00:00.000	Henry	Daley	Residential	BROOKLYN	11249	Not an Ejectment	Possession	40.717722	-73.962014
479082	483-485 COURTLANDT A VENUE	4C	2018-08-07T00:00:00.000	Danny	Weinheim	Residential	BRONX	10451	Not an Ejectment	Possession	40.815073	-73.920228
109031	1191 BROADWAY	UNIT 2	2024-05-10T00:00:00.000	Justin	Grossman	Commercial	BROOKLYN	11221	Not an Ejectment	Possession	40.692442	-73.927219

In NYC evictions are executed by Marshals (independent public officers appointed by the Mayor to enforce Civil Court orders).

The Borough-Block-Lot (BBL) Identifier



Image source: NYCityMap

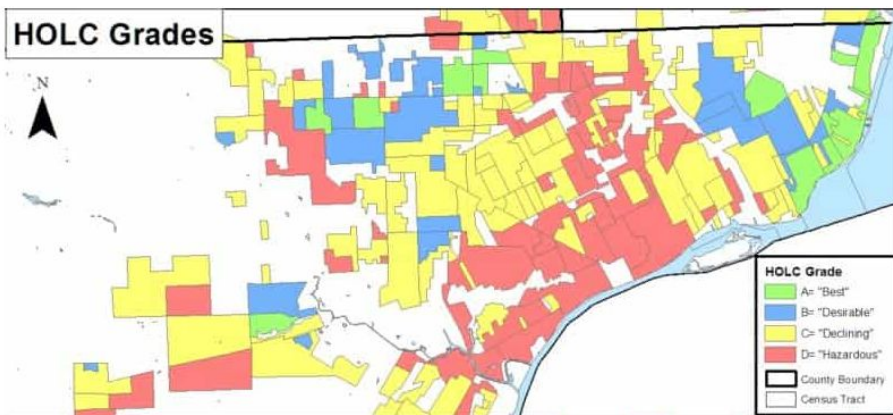
“Borough, Block, and Lot (also called Borough/Block/Lot or BBL) is the parcel number system used to identify each unit of real estate in New York City for numerous city purposes”

— Wikipedia

**available within MapPLUTO, abbreviation of primary land use tax output, which is NYC Department of City Planning’s geospatial tax lot data*

NCRC (National Community Reinvestment Coalition) Redlining Dataset

- Helen Meier and Bruce Mitchell mapped the Historical Redlining Indicator (HRI) onto census tracts
- Home Owners' Loan Corporation (HOLC)
 - the U.S. federal agency that graded mortgage investment risk of neighborhoods across the U.S. between 1935 and 1940
- Historical data
 - a grade of "A" indicates low risk
 - each HOLC risk category is assigned a numeric score
 - 1 for "A" grade, 2 for "B" grade, 3 for "C" grade, and 4 for "D" grade
 - each tract's score is gotten by an areal proration from the HOLC zones



Detroit, Michigan

Image source: <https://ncrc.org/redlining-score/>

Discussion: Evictions vs. Redlining Index

